

Students will name, measure, and categorize angles.

Do Now

1) List all the things that you know about angles.

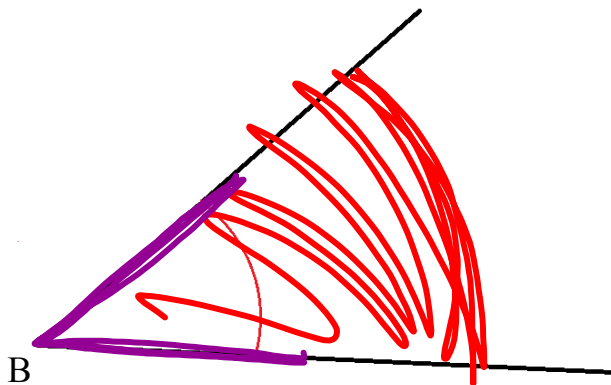
- 3 diff types (right, obtuse, acute)
- can be $\cong$. (straight, reflex, circle)

2) How would you define "angle" in your own words?

- 2 lines connect to make an
 $\angle$ from 0 - 360.
- Area b/w 2 lines connected
 at same point

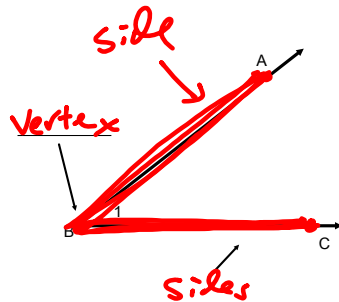
Do Next

3) Which angle is bigger? Justify your response.



Defining and Naming Angles

In Geometry, an angle is a Figure (shape) formed when two sides meet at a common point, called a Vertex. We use the symbol $\angle$ to denote an angle.



An angle can be named in 3 different ways:

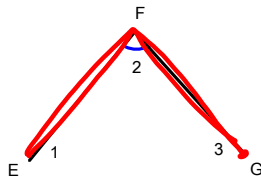
1) By its vertex: $\angle B$

2) By its number: $\angle 1$

3) By its sides & vertex: $\angle ABC$ or $\angle CBA$

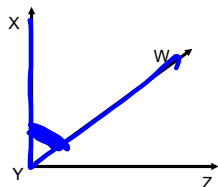
Note: The vertex must be in middle!

4) Name the angle shown in the diagram below in three different ways.



$\angle GFE$ $\angle EFG$
 $\angle F$ $\angle 2$

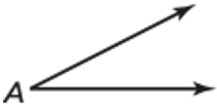
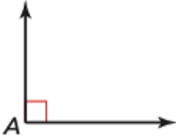
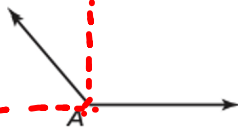
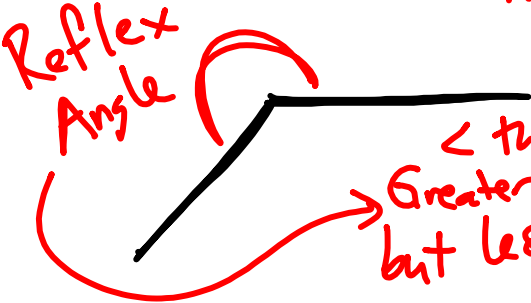
5) On a recent Geometry Regents exam, a student labeled the marked angle as $\angle Y$. Explain why they got points taken off.



$\angle Y$? $\angle Y$ could mean 3 different $\angle$ s.

Correction: Call it $\angle XYW$.

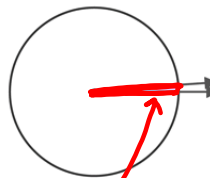
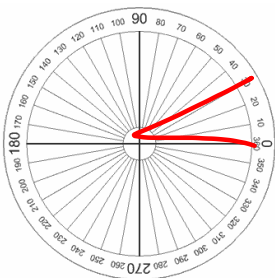
Types of Angles

Acute Angles	Right Angle	Obtuse Angle	Straight Angle
			
$\angle$ that is less than 90°	$\angle$ that is 90°	$\angle$ that is greater than 90° (and less than 180°)	$\angle$ that is 180° ; straight line.
Reflex Angle  $\angle$ that is Greater than 180° but less than 360°.			

Measuring Angles

Angles are measured using units called degrees. Alternatively, they are measured using units called radians.

A degree is defined as $1/360$ th of a rotation around a circle.



$$\frac{1}{360} = 1^\circ$$

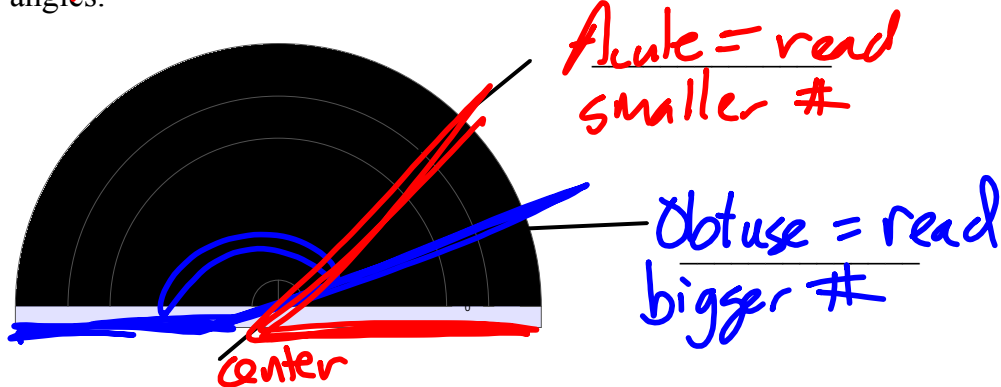
one slice of pizza that has 360.

Why do you think there are 360 degrees in a circle?

- Matches ≈ 365 days in year
- 360 has many divisors (Base 10 # systems)

Using a Protractor

A protractor is a tool that can be used to measure the degrees of angles.

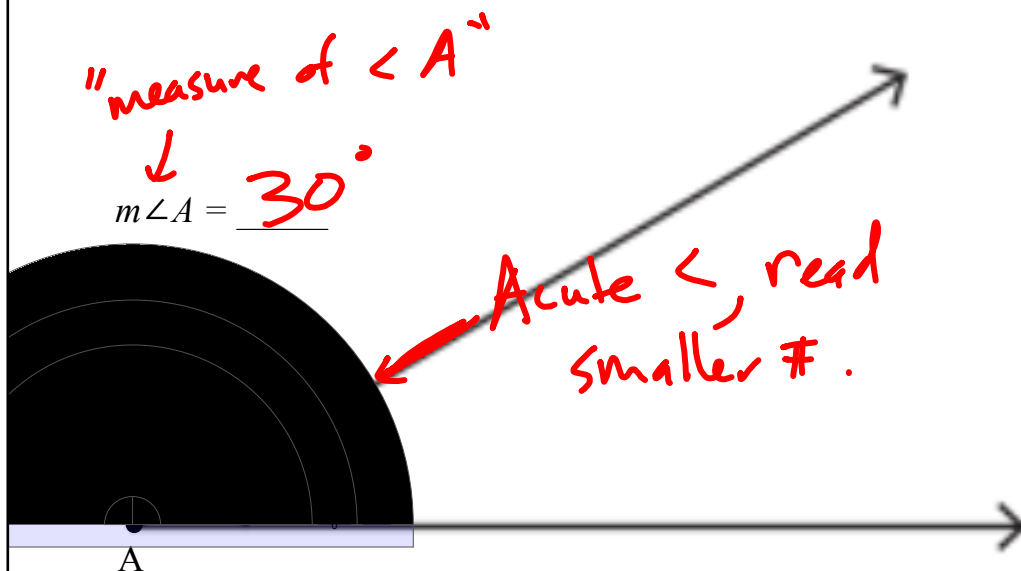


To use a protractor:

- Step 1: Place the center of the protractor's bottom (usually a small hole or + on the protractor) on the vertex.
- Step 2: Align 0° with one of the sides of your angle.
- Step 3: Read the inner (for acute angles) or outer (for obtuse angles) number to find the angle measure.

6) Use your protractor to determine the angle measure of angle A below.

[Pro tip: you might need to use the straight edge of your compass to extend the sides of your angle.]



7) Use your protractor to determine $m\angle B$.

[Pro tip: you might need to use the straight edge of your compass to extend the sides of your angle.]

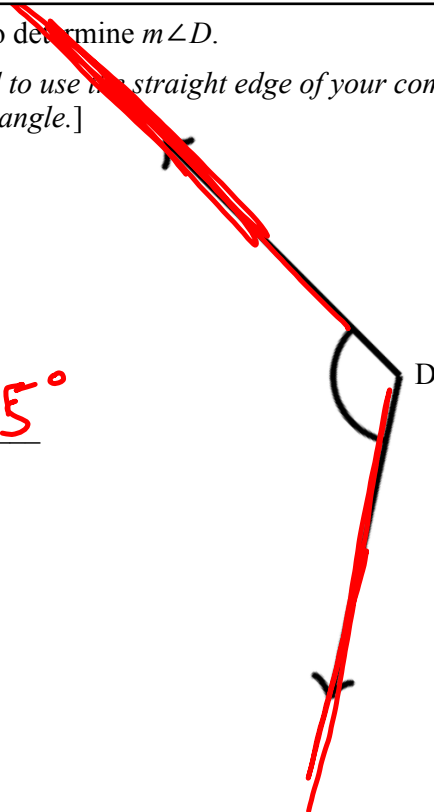


$$m\angle B = \underline{30^\circ}$$

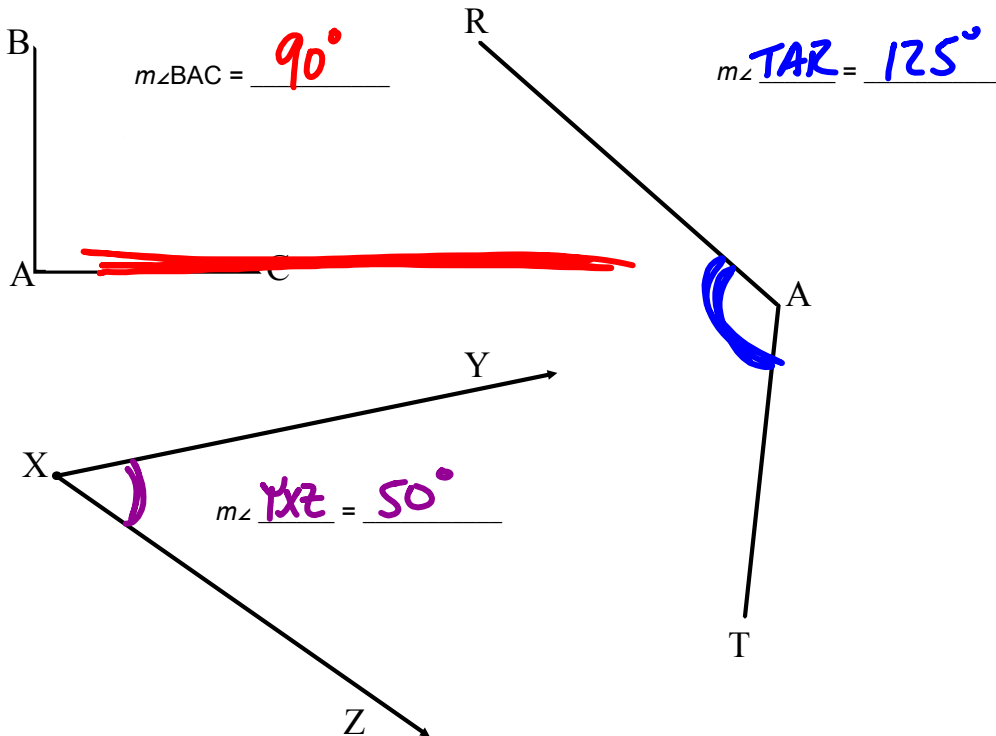
8) Use your protractor to determine $m\angle D$.

[Pro tip: you might need to use the straight edge of your compass to extend the sides of your angle.]

$$m\angle D = \underline{125^\circ}$$



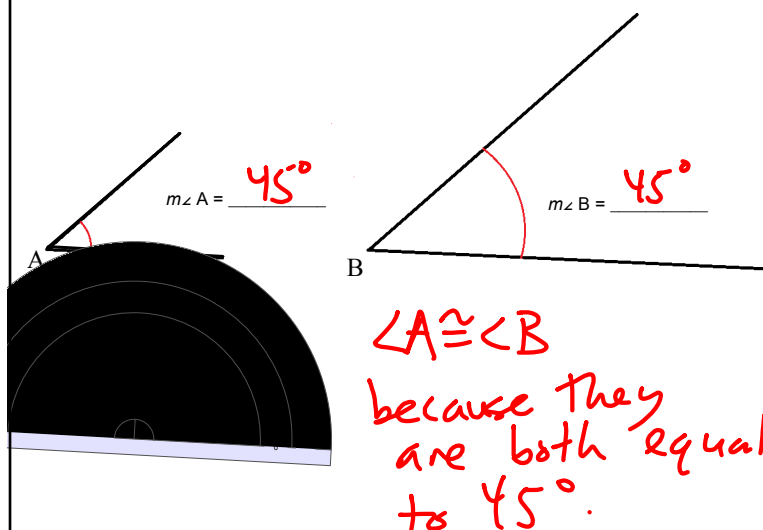
9) Use a protractor to measure each of the following angles.



Angle Congruence

Two angles are said to be congruent ($\cong$) if they have the same angle measure.

10) Are the two angle below congruent? Justify your response.



You Try

Find the measures of each of the following angles, then determine if they are acute, obtuse, straight, or right.

11) $\angle GHK$

$m\angle GHK \approx 125^\circ$
obtuse

12) $\angle JHL$

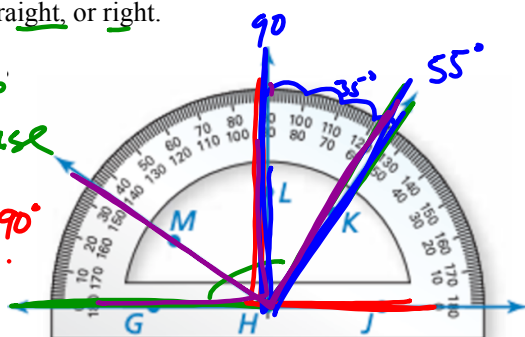
$m\angle JHL = 90^\circ$
rt.

13) $\angle LHK$

$m\angle LHK = 35^\circ$

14) Identify two different congruent angles.

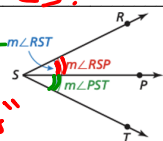
$\angle KHM \cong \angle LHG$



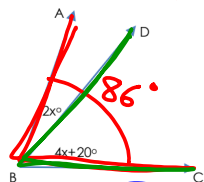
Angle Addition Postulate

The Angle Addition Postulate states that a big $\angle$ is the sum of its smaller $\angle$ s.

$m\angle RST = m\angle RSP + m\angle PST$
"whole = sum of its parts"



15) If $m\angle ABC = 86^\circ$, determine the $m\angle DBC$.



$86 = 2x + 4x + 20$

$86 = 6x + 20$
 -20

$\frac{66}{6} = \frac{6x}{6}$

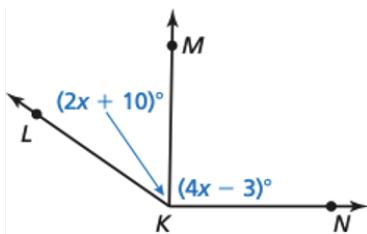
$11 = x$

$m\angle DBC =$
 $4x + 20$
 $4(11) + 20$
 64°

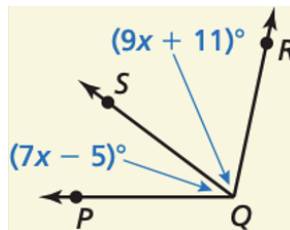
Practice

Find the measure of the unknown angles:

16) $m\angle LKN = 145^\circ$



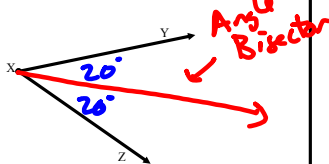
17) $m\angle PQR = 102^\circ$



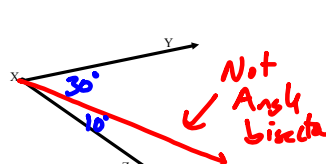
Angle Bisectors

An **angle bisector** is a line/segment/ray that splits an $\angle$ into 2 congruent $\angle$ s.

Example:



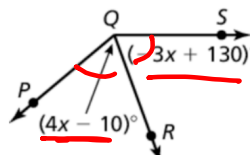
Non-Example:



Ex: Ray QR bisects $\angle PQS$. Find $m\angle PQR$ and $m\angle POS$.

$\angle$ bisector!!!!

$$4x - 10 = -3x + 130$$



18) Ray BD bisects $\angle ABC$. Find $m\angle ABC$ and $m\angle DBC$ if $m\angle ABD = 43^\circ$.